

One Interval, Two Threshold Claims

AP Statistics · Unit 3 · Topic 3.4 · B: 2026-12-07 · E: 2026-12-09

[STAR] TODAY'S PROBLEM

Suppose a statewide activities association has 10,000 member students and is considering requiring reusable food containers at its tournaments. In a simple random sample of 300 member students, 204 favor the requirement. A sponsor claims that at least 65% of all member students favor it; an event organizer makes the more modest claim that more than 60% favor it.

Kai: “The sample has $204/300 = 68\%$ in favor, above both thresholds, so it supports both claims.”

Mira: “The point estimate alone cannot settle both claims. We should construct a confidence interval and compare all of its plausible values with each threshold.”

Sample counts

Group	N	n	Favor	Other
Members	10,000	300	204	96

FIRST TAKE — before we discuss (5 min)

Do the data appear to support both claims, only one, or neither? Cite one number and commit before calculating the interval.

[BOOK] LET THE WHOLE INTERVAL SET THE BOUND (keep on the page)

For a one-sample z -interval for p , check random sampling, $n \leq 0.10N$, and at least 10 observed successes and failures. Then use $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ for 95% confidence. Interpret: “We are 95% confident that the interval captures the population proportion [in context].” A claim $p > c$ is supported only when the whole interval lies above c ; if c is inside the interval, plausible values lie on both sides.

Work (a)–(c) from the rules box:

DOK 1 (a) Define the population proportion p and calculate the sample proportion $\hat{p}$.

DOK 2 (b) Verify the conditions, construct a 95 percent confidence interval for p , and interpret the interval in the association's context.

DOK 3 (c) In 3–4 sentences, decide which threshold claim the interval supports and whose reasoning is better. Compare the lower endpoint with 0.60 and 0.65. Explain why a plausible 0.65 is different from establishing at least 0.65.

Frame: The interval supports the claim _____ because its lower endpoint, _____, is _____.

Frame: Although 0.65 is _____, it does not establish _____; Kai's argument would make readers think _____.

EXIT — turn this in

One thing the rules box changed about my first take: _____